

AIT Programme — Complete Rulebook

Categorisation, Coaching, Performance, Final Marks, and Advanced Ranking

This rulebook is in two parts. Part A defines the core programme: how students are categorised after each test, how coaches are allotted, how coach performance is measured, and how final-exam marks are computed. Part B defines an advanced ranking system that builds on Part A to determine Rank 1 and Rank 2, using the previously unused 10th and 12th percentages and a self-referential coach-quality loop. All three data files (Students List, AIT Marks List, Final Exam Attendance) are used unchanged.

Conventions used throughout: $n\%$ means a factor of $(1 + n/100)$ for a bonus or $(1 - n/100)$ for a penalty; bounds are stated explicitly as inclusive ($\geq$, $\leq$) or exclusive ($>$, $<$); rounding is applied only where stated.

PART A — Core Programme Rules

A1. Student Categorisation

After each AIT, every student is placed in a category based on their running average score. Let a be the average of all AIT marks scored up to and including the current test (after AIT-1 that is the AIT-1 mark; after AIT-2 the mean of AIT-1 and 2; and so on). Each paper is out of 360.

Category	Condition on average a
a1	$a \geq 300$
a2	$250 \leq a < 300$
a3	$200 \leq a < 250$
a4	$a < 200$

A2. Coach Allotment

Students are split into four batches of 200. A student's batch is fixed, but the allotted coach depends on both batch and current category, so the coach can change between tests as the category changes:

Batch	Category a1	Category a2	Category a3	Category a4
Batch 1	Coach A	Coach B	Coach C	Coach D
Batch 2	Coach B	Coach C	Coach D	Coach A
Batch 3	Coach C	Coach D	Coach A	Coach B
Batch 4	Coach D	Coach A	Coach B	Coach C

A3. Measuring Coach Performance

A3.1 Per-student improvement between two AITs

For each pair of consecutive tests, a student's percentage improvement is the relative change in marks:

$$\% \text{ improvement} = (\text{later mark} - \text{earlier mark}) \div \text{earlier mark} \times 100$$

This yields three values per student (AIT-1→2, 2→3, 3→4); a value is negative when the score falls.

A3.2 Attributing improvement to a coach

Each transition's improvement is credited to the coach the student was assigned after the earlier test — the coach who prepared them for the next test. So the AIT-1→2 improvement is credited to the coach allotted after AIT-1.

A3.3 Average improvement per coach, per transition

For each transition, a coach's average is the total improvement of all students that coach handled in that transition divided by the number of those students:

$$\text{Average (coach, transition)} = \sum \text{student improvements} \div \text{number of students coached}$$

A3.4 Overall coach performance

A coach's overall performance, written `base_coach`, is the simple average of their three per-transition averages, weighting each transition equally:

$$\text{base_coach} = (\text{Avg}(1 \rightarrow 2) + \text{Avg}(2 \rightarrow 3) + \text{Avg}(3 \rightarrow 4)) \div 3$$

A4. Final Exam Marks (base programme)

After AIT-4 each student's final category and coach are allotted using the average of all four AITs. The coach then provides an improvement of p percent over the AIT-4 mark.

A4.1 The improvement percentage p

$$p = p_0 + 4b - c$$

- p_0 = `base_coach` for the student's AIT-4 coach (Section A3.4).
- b = number of transitions in which marks improved versus the previous AIT (out of the three transitions).
- c = number of transitions in which marks decreased versus the previous AIT.

Since each transition is an improvement or a decrease, $b + c = 3$ for every student.

A4.2 The final mark

$$\text{Final mark} = \text{AIT-4 mark} \times (1 + p / 100)$$

A4.3 Attendance adjustment

The mark is then adjusted for exam-day attendance:

- Present: used as-is.
- Late by n minutes: reduced by n percent, i.e. multiplied by $(1 - n/100)$.
- Absent: no mark is calculated; the student is excluded from any ranking.

Adjusted final mark = Final mark $\times$ (1 - n / 100), n = minutes late (0 if present)

Worked example: AIT-4 = 286, coach overall = 3.04%, b = 2, c = 1, present $\Rightarrow p = 3.04 + 8 - 1 = 10.04\%$, final = $286 \times 1.1004 = 314.71$.

PART B — Advanced Final Ranking

Part B determines Rank 1 and Rank 2 across the cohort. It reuses the categories and coach table of Part A and the base_coach figures of Section A3.4, but generalises the final mark of Section A4: the static p_0 is replaced by an iteratively-solved coach quality Q , and the coach used is the one obtained after a capacity-limited allocation. The stages are deeply interdependent — eligibility, capacity allocation, within-batch normalization and a self-referential loop all feed one another — so no student's result can be computed in isolation; the whole eligible cohort must be solved together, by iteration, until it stabilises.

B0. Academic Index (AI)

Using the 10th and 12th percentages from the Students List:

$$AI_0 = 0.4 \times (10\text{th } \%) + 0.6 \times (12\text{th } \%)$$

Decline adjustment: if 12th % is strictly less than 10th %, multiply by 0.95; otherwise leave unchanged. Round to two decimals after the multiplier.

$$AI = AI_0 \times 0.95 \quad \text{if } 12\text{th } \% < 10\text{th } \% ; \quad AI = AI_0 \quad \text{otherwise}$$

B1. Eligibility Gate

A student is eligible for ranking only if ALL of the following hold; otherwise they are excluded from ranking (but still coached in B2).

- E1: the student is NOT Absent.
- E2: $AI \geq 66.00$ (exactly 66.00 is eligible — inclusive bound).
- E3: at most one AIT-to-AIT decrease ($c \leq 1$) — UNLESS $AI \geq 85$, in which case E3 is waived and the student is eligible regardless of c .

B2. Coach Allocation with Capacity

Every student (eligible or not) first receives a provisional coach exactly as in Part A (four-AIT average → category → batch-and-category table). Then a capacity rule applies: each coach holds at most 205 students. Students claim their provisional coach one at a time in strict priority order:

- highest AI first;
- ties broken by higher AIT-4 mark;
- further ties broken by lower Roll No.

If a student's provisional coach is full, they overflow to the next coach in the rotation $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$, continuing around until one has space. The coach a student ends up with is their FINAL coach, used everywhere downstream.

B3. Within-Batch Normalization

For each ELIGIBLE student compute a recency-weighted AIT strength (later tests weigh more):

$$S = (1 \times AIT1 + 2 \times AIT2 + 3 \times AIT3 + 4 \times AIT4) / 10$$

Then standardise within the student's batch, using the mean and population standard deviation over ELIGIBLE students in that batch only:

$$Z = (S - \mu_{\text{batch}}) / \sigma_{\text{batch}}$$

B4. Iterative Coach-Quality Loop

Coach quality Q is defined self-referentially in terms of how that coach's own eligible students actually perform on the final exam, so it must be solved by iteration rather than algebra. Each eligible student's bonus uses their FINAL coach (B2) and their own b , c :

$$p = Q_{\text{(final coach)}} + 4b - c$$

The final mark applies $p\%$ above AIT-4, then the late penalty, rounding only at the end:

$$F = \text{ROUND}(\text{AIT4} \times (1 + p/100) \times (1 - n/100) , 2)$$

Each student's realised final-exam percentage gain is:

$$g = (F - \text{AIT4}) / \text{AIT4} \times 100$$

Coach quality is then the blend of the historical base with the median realised gain of that coach's eligible students:

$$Q_{\text{coach}} = 0.5 \times \text{base_coach} + 0.5 \times \text{MEDIAN}(g \text{ of that coach's eligible students })$$

B5. Composite Score

The quantity that is ranked is computed from the converged final marks:

$$\text{Composite} = 0.5 \times (50 + 10 \times Z) + 0.3 \times F + 0.2 \times \text{AI}$$

Here $(50 + 10 \times Z)$ re-scales the within-batch Z-score, F is the late-adjusted final mark from B4, and AI is the Academic Index from B0.

B6. Ranking, Rank 1 and Rank 2

Rank all ELIGIBLE students by final Composite descending, breaking ties in order:

- higher late-adjusted final mark F ;
- then lower c (fewer decreases);
- then higher 12th %;
- then lower Roll No.

Rank 1 is the highest Composite; Rank 2 is the next. Absent students are not ranked.